

National Institute of Technology Rourkela

Department of Computer Science and Engineering

Mid-Term Examination (Feb 2024)

Deep Learning (CS6510), B.Tech. 8th Semester

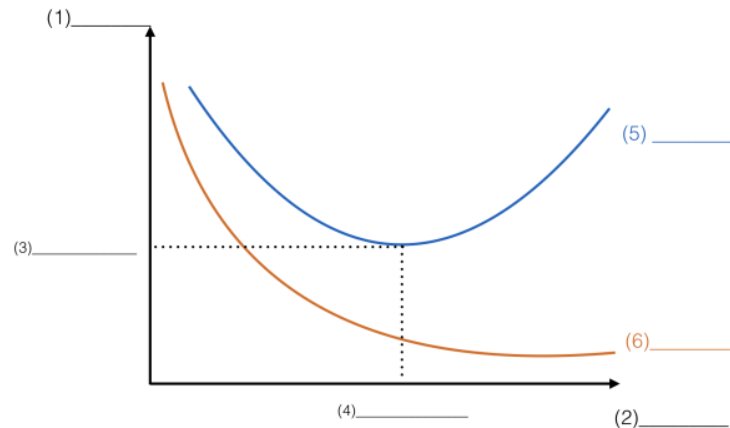


Time: 120 Minutes

Max. Marks: 30

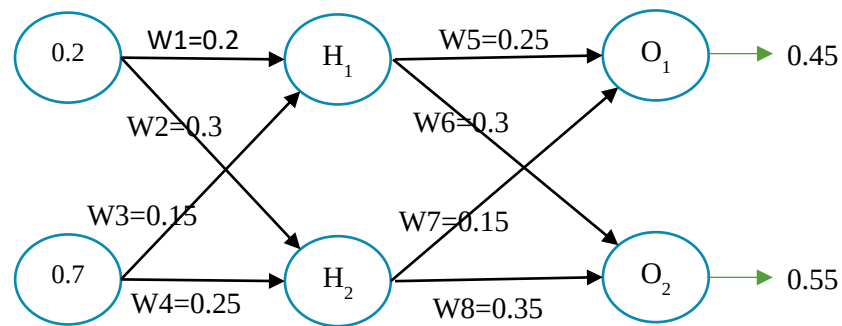
Instruction: 1. In case of any doubt, write your assumptions, write it clearly and continue.
2. There are 09 questions printed on the paper. Attempt all the questions.

- Q1. A 1-hidden layer neural network with k neurons, n inputs and m outputs with bias at each neuron will have how many total parameters. **01**
- Q2. Consider a fully connected network with n inputs $x_1, x_2, \dots, x_n$. Suppose there is a single hidden layers with k neurons having f as activation function and b_j as the bias of j th hidden neuron. Further, the output layer contains a single neuron with g as activation function and B as the bias. Assume that the weights between inputs to hidden layer is denoted by w and weights between hidden to output layer is denoted as v . Write down the output y of the network as a function of x , weights and biases. **02**
- Q3. You come across a nonlinear function that passes 4 if its input is nonnegative, else evaluates to -2, i.e. $f(x) = \begin{cases} 4 & \text{if } x \geq 0 \\ -2 & \text{Otherwise} \end{cases}$ **02**
A friend recommends you to use this non-linearity in your deep neural network with the Adam optimizer. Would you follow his advice? Why or why not?
- Q4. You want to use the figure below to explain the concept of early stopping to a friend. Fill-in the blanks. (1) and (2) describe the axes. (3) and (4) describe values on the vertical and horizontal axis. (5) and (6) describe the curves. Be precise. **03**



- Q5. What is the difference between gradient descent, stochastic gradient descent and mini-batch stochastic gradient descent? Given a dataset having 10020 input instances. How many times the weights will be updated in one epoch if Gradient Descent, Stochastic Gradient Descent and mini-batch stochastic gradient descent with 128 batch size are separately used? **05**

- Q6. If the sets P and N are finite and linearly separable, prove that the perceptron learning algorithm updates the weight vector w_t a finite number of times such that the two sets are separated. **04**
- Q7. Prove that moving in the opposite direction to the function Gradient minimizes the function. Explain the Backpropagation algorithm using gradient descent update rule. **04**
- Q8. What problem occurs if the learning rate is too high or too low? Why adaptive learning rate with respect to cumulative sum of gradients are used in Adagrad optimizer? Explain how the problem of Adagrad is overcome in RMSProp by mentioning the weight update equations? **05**
- Q9. Consider the following neural network architecture with sigmoid activation function at every neuron and given weights as initial weights, inputs and outputs. Compute the value of W_4 , W_7 and W_8 after 1 epoch of backpropagation algorithm using stochastic gradient descent weight updating rule with learning rate is 0.1. **04**



*****End of Question Paper*****